

Adaptive Sample Size Modification in Clinical Trials:

Start Small then Ask for More?

Workshop with R programs

Christopher Jennison

Department of Mathematical Sciences,

University of Bath, UK

<http://people.bath.ac.uk/mascj>

Roche, Welwyn Garden City

May 2013

1. Choosing the sample size for a trial

We denote by θ the effect size of a new treatment, i.e., the difference in mean response between the new treatment and the control.

Sample size is determined by:

Type I error rate α , and

Treatment effect size $\theta = \Delta$ at which power $1 - \beta$ is to be achieved.

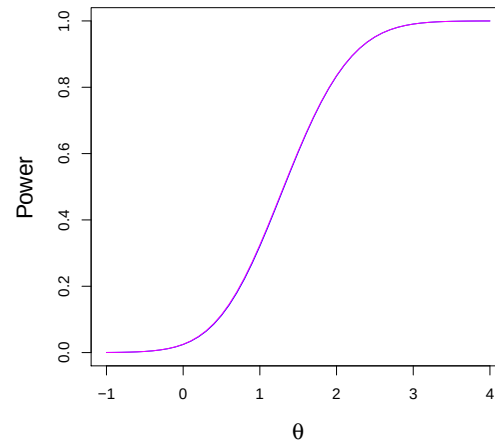
When investigators consider the choice of Δ and its impact on sample size, should they:

Play safe and use the minimum effect of interest Δ_1 , or

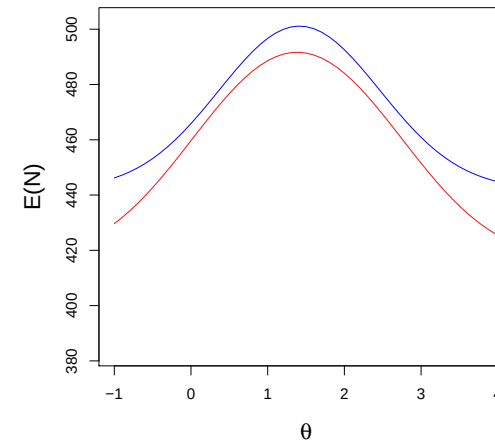
Be optimistic and use an effect size $\Delta_2 > \Delta_1$?

Designing a clinical trial

Power curve



$E_{\theta}(N)$ curves



All designs, *including adaptive procedures*, have overall power curves.

Designs with similar power curves can be compared in terms of their average sample size functions, $E_{\theta}(N)$.

Values of θ may be specified under which early stopping is most desirable.

But, if two designs have the same power and Design A has *uniformly lower* $E_{\theta}(N)$ than Design B, then Design A should be preferred.

2. Mehta & Pocock's Example

We shall consider MP's Example 1, a Phase 3 trial for schizophrenia, in which a new drug is to be compared to an active comparator.

The efficacy endpoint is improvement in the Negative Symptoms Assessment score from baseline to week 26.

Responses are

$$Y_{Bi} \sim N(\mu_B, \sigma^2), \quad i = 1, 2, \dots, \text{ on the new treatment,}$$

$$Y_{Ai} \sim N(\mu_A, \sigma^2), \quad i = 1, 2, \dots, \text{ on the comparator treatment.}$$

where $\sigma^2 = 7.5^2$.

The treatment effect is

$$\theta = \mu_B - \mu_A.$$

and we estimate θ by

$$\hat{\theta} = \hat{\mu}_B - \hat{\mu}_A = \bar{Y}_B - \bar{Y}_A.$$

Mehta & Pocock's Example

The initial plan is for a total of $n_2 = 442$ patients, 221 on each treatment.

In testing $H_0: \theta \leq 0$ vs $\theta > 0$, the final analysis will reject H_0 if Z_2

$$Z_2 = \frac{\hat{\theta}(n_2)}{\sqrt{\{4\sigma^2/n_2\}}} > 1.96.$$

This design and analysis gives type I error rate 0.025 and power 0.8 at $\theta = 2$.

Higher power, e.g., power 0.8 at $\theta = 1.6$, would be desirable.

We shall investigate how this may be achieved.

(1) Running R: Initial set up

You can run R in a particular directory on your computer, making it easy to use files in this directory and save your results there.

1. Create a directory for this workshop
2. Copy all the files with a .R suffix from the memory stick to this directory.
3. Start running R and navigate to your “workshop” directory

After starting R, click on **File**, then on **Change dir**

Navigate to find the directory you have created and double click on its name.

As a check, you can find the current directory by typing

```
> getwd( )
```

The R files from the memory stick should now appear as options when you try to load a file or open a script.

(1) Running R: **fixed_design**

To use the R routine **fixed_design.R**, you need to load this into your workspace.

To do this, click on **File**, then on **Source R Code**.

Find the file **fixed_design.R** and double click on this to load the code.

If you then type

```
> ls()
```

You should see the function **fixed_design** listed as being present in the workspace.

Type

```
> fixed_design(442)
```

to run this function with a sample size of 442.

There should be a plot of a power curve and a plot of the sample size function (which is equal to 442 for all θ) — if these plots are overlaid, move one aside to see the other.

(1) Running R: **fixed_design**

A sample size of 442 gives power a little greater than 0.6 at $\theta = 1.6$.

Experiment to see how large a sample size is needed to raise this power to 0.8.

Typing

```
> nf = c(442, 500, 550)
```

creates the vector variable **nf** containing three values.

Use this in the command

```
> fixed_design(nf)
```

Observe that R functions can take either scalars or vectors as arguments.

(1) Running R: Using a script

I have prepared a “script” of commands that you can use when calling the function `fixed_design`.

To open this script, click on **File** and **Open Script . . .**, then find `run_fixed_design.R` and double click on this to open the script.

If you highlight commands in the script and type Control R, these commands will run in the R Console.

You can run all the commands in a script by typing Control A, followed by Control R.

Try editing the script and re-running the commands.

If you want to see how the function `fixed_design.R`, is defined, open this as a script and take a look.

(1) Notes on using Scripts

Having an open script file is a useful way to develop or modify a set of R commands.

If all you want to do is run the commands, you can simply load the file (click on [File](#), then on [Source R Code, etc.](#)) and the full set of commands will be implemented.

With an open script, you can choose to run a few commands at a time, edit some lines and run them again, and so forth.

You can save changes to a script by clicking on [File](#) and [Save](#) while the script window is active.

I suggest that

- (a) You keep scripts I have provided unchanged, by using the [Save as . . .](#) option and giving a new name.
- (b) You close each script as you finish using it — it is easy enough to open it again if you need to.

3. The MP design: Increasing sample size adaptively

In Mehta & Pocock's example, we are told that the sponsors of the trial are willing to commit a sample size of 442 patients.

They will allow this number to be increased if interim results are "promising".

At an interim analysis after $n_1 = 208$ responses have been observed, we have

$$\hat{\theta}_1(n_1) = \bar{Y}_B(n_1) - \bar{Y}_A(n_1)$$

and

$$Z_1 = \frac{\hat{\theta}_1(n_1)}{\sqrt{\{4\sigma^2/n_1\}}}.$$

At the time of this analysis, a further 208 subjects will have been treated for less than 26 weeks. Their responses will be observed in due course.

As recruitment continues, we use the value of Z_1 , or equivalently $\hat{\theta}_1(n_1)$, in choosing a new total sample size, up to a maximum of 884.

Increasing the sample size

Definition

The conditional power $CP_{\theta}(z_1)$ is the probability the final test, with $n_2 = 442$ observations, rejects H_0 , given $Z_1 = z_1$ and effect size θ ,

$$CP_{\theta}(z_1) = P_{\theta}\{Z_2 > 1.96 \mid Z_1 = z_1\}.$$

MP's adaptive design is based on conditional power under $\theta = \hat{\theta}_1$.

They divide the range of z_1 into three regions:

<i>Favourable</i>	$CP_{\hat{\theta}_1}(z_1) \geq 0.8$	<i>Continue to $n_2 = 442$,</i>
<i>Promising</i>	$0.365 \leq CP_{\hat{\theta}_1}(z_1) < 0.8$	<i>Increase n_2,</i>
<i>Unfavourable</i>	$CP_{\hat{\theta}_1}(z_1) < 0.365$	<i>Continue to $n_2 = 442$.</i>

When increasing sample size in the promising zone, the final test of H_0 must protect the type I error rate at level α .

The Chen, DeMets & Lan method

References:

Chen, DeMets & Lan, *Statistics in Medicine* (2004),

Gao, Ware & Mehta, *J. Biopharmaceutical Statistics* (2008).

Suppose at interim analysis 1, the final sample size is increased to $n_2^* > n_2$ and a final test is carried out without adjustment for this adaptation.

Thus, H_0 is rejected if

$$Z_2(n_2^*) = \frac{\hat{\theta}(n_2^*)}{\sqrt{\{4\sigma^2/n_2^*\}}} > 1.96.$$

Chen, DeMets & Lan (CDL) show that if n_2 is only increased when

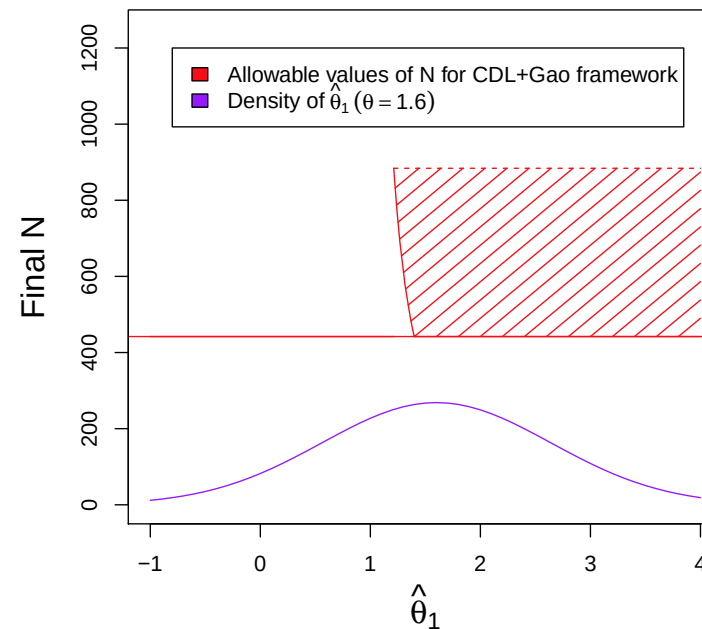
$$CP_{\hat{\theta}_1}(z_1) > 0.5,$$

then the type I error probability will not increase.

(In general, changes to sample size may increase or decrease the type I error rate.)

Gao's extension of the CDL method

Gao et al. extended the CDL method to lower values of $\hat{\theta}_1$, as long as a sufficiently high value is chosen for the final sample size, n_2^* .

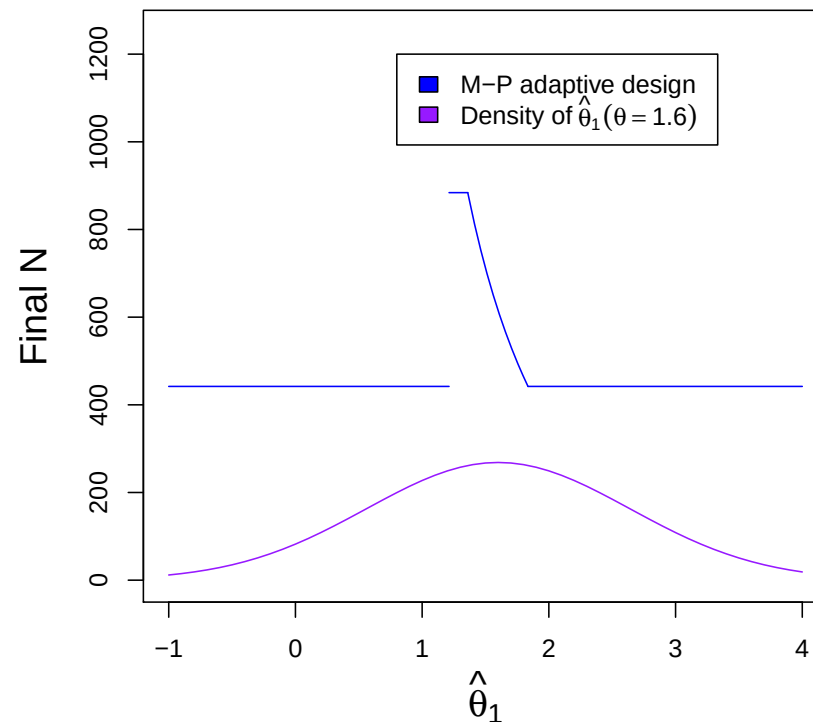


With an upper limit of $n_2^* = 884$, the final sample sizes permitted by the CDL+Gao approach are as shown in the figure.

Now n_2 can be increased when $CP_{\hat{\theta}_1}(z_1)$ is as low as 0.365.

The MP design

In their “promising zone”, MP increase n_2 to achieve conditional power 0.8 under $\theta = \hat{\theta}_1$, truncating this value to 884 if it is larger than that.



We shall investigate this design and variations on it where we change the targets for the conditional power and set different limits on the maximum sample size.

(2) Running R: **mp_design**

Load the R functions in the file **mp_design.R**, into your workspace by clicking on **File**, then on **Source R Code**. When you have found the file **mp_design.R**, double click on this to load the code. If you then type

```
> ls()
```

You should see the function **mp_design** listed as being present in the workspace.

Run this function with a maximum sample size of 844 and a target of 0.8 for conditional power under $\theta = \hat{\theta}_1$ by typing

```
> mp_design(884, 0.8)
```

Three plots should appear:

1. The MP sample size rule.
2. Power curves of the MP design and a fixed sample trial with $N = 442$.
3. $E_{\theta}(N)$ curves for these designs and for a fixed sample trial with the same power as the MP design.

(2) Running R: **mp_design**

Study the plots produced by the command

```
> mp_design(884, 0.8)
```

Comparison of the sample size modification rule with the distribution of $\hat{\theta}_1$ under $\theta = 1.6$ shows that increases in n_2 occur in a region of quite small probability.

The gain in power is modest. A fixed sample size design with $N = 492$ gives an almost identical power curve to the Mehta-Pocock design.

Experiment with other values for the maximum sample size and target conditional power. For example, if you wish to achieve conditional power under $\theta = \hat{\theta}_1$ of 0.9 and are willing to allow a total sample size of 900, type

```
> mp_design(900, 0.9)
```

You can use the script **run_mp_design.R** if you wish.

What are your conclusions?

(3) Can we improve the trial design within the MP framework?

Comparisons with fixed sample designs and group sequential designs show the MP design to have high $E_{\theta}(N)$ for its achieved power.

(And these group sequential designs are wasteful as they ignore data from “pipeline” subjects.)

The principles underlying the MP design appear to be sound: observations should be added when they will give the most benefit.

Why are the results not so good?

What would you suggest to improve this design?

(3) Improving a two-stage, adaptive trial design

Some possible modifications are:

- Use a combination test to protect the type I error rate while permitting sample size increase (or decrease) for **any** $\hat{\theta}_1$.
- Aim for a specified conditional power
at a fixed effect size $\theta = \delta$,
or integrated over a posterior distribution of θ (for some prior).
- Stop for futility (or only observe “pipeline” subjects) if conditional power is low.
- Consider costs (for sample size) and benefits (higher power) explicitly, using a consistent rate of exchange between these.
- Create a design which has optimal overall properties — as low an $E_\theta(N)$ curve as possible for the achieved power curve.

4. A decision theory approach to cost-benefit analysis

Suppose we can elicit the following information:

There is a probability P_{sup} that the new treatment is superior to control.

If the treatment is effective, investigators expect the effect size to be θ_0 .

The cost of each patient in the Phase III clinical trial is $\mathcal{L} C$.

If a test of $H_0: \theta \leq 0$ is rejected in a 1-sided test at level $\alpha = 0.025$,
the drug succeeds and the financial return is $\mathcal{L} G C$.

Then, the expected gain, in pounds, from a trial with N patients is

$$P_{sup} \times \text{Power}(N, \theta_0) \times G \times C - N \times C.$$

where $\text{Power}(N, \theta_0) = P_{\theta=\theta_0}\{\text{Reject } H_0: \theta \leq 0\}$ with a sample size of N .

A decision theory approach to cost-benefit analysis

We have the expected gain, in pounds, from a trial with N patients given by

$$\begin{aligned} E(\text{Gain}) &= P_{sup} \text{Power}(N, \theta_0) G C - N C \\ &= P_{sup} G C \left(\text{Power}(N, \theta_0) - \frac{1}{P_{sup} G} N \right) \\ &= P_{sup} G C (\text{Power}(N, \theta_0) - \gamma N), \end{aligned} \tag{1}$$

where

$$\gamma = \frac{1}{P_{sup} G}.$$

The decision theory solution for the optimal sample size is the value of N that maximises (1).

(4) Running R: **gain_fixed**

Load the R function in the file **gain_fixed.R** into your workspace.

Open the script **run_gain_fixed.R** and use the commands there to run **gain_fixed**.

You should see a plot of $Power(N, \theta_0)$ as N varies between zero and $N_{\max}$.

Below this is a plot of the “combined objective”, $Power(N, \theta_0) - (1/\gamma) N$, which is a multiple of the expression we have just seen for $E(Gain)$.

You can replace α , θ_0 , $P_{\sup}$ and G with your own values and run the script again.

In some cases, it helps to change $N_{\max}$ in order to see more of the power curve.

What values of the variables in the decision theory model are needed to give the initial choice of sample size, $N_{\max} = 442$, in MP's example?

Do these values seem reasonable?

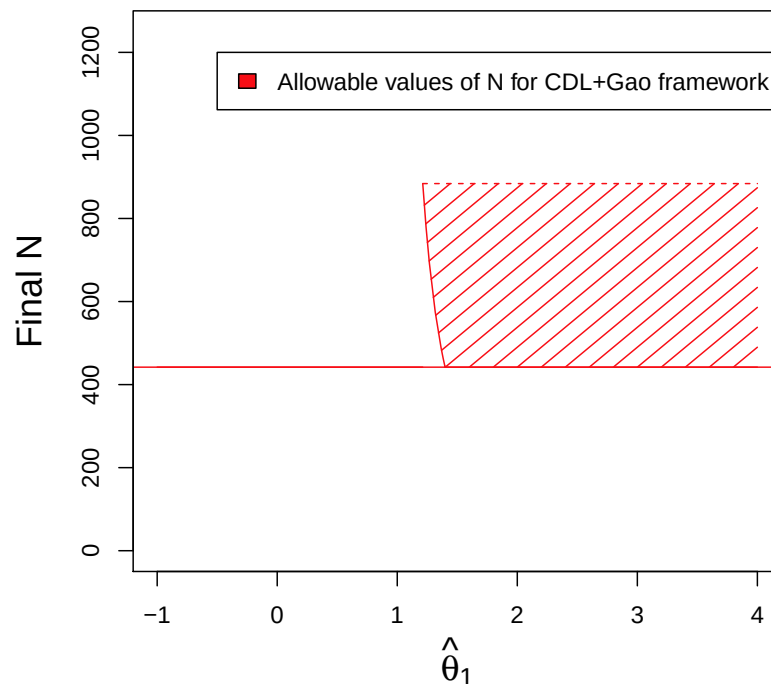
Which aspects of this problem formulation do you find the least realistic?

5. Deriving efficient sample size rules in the MP framework

We stay with MP's example and retain many elements of their design.

The interim analysis takes place after 208 observed responses.

A final sample size n_2^* is chosen based on $\hat{\theta}_1$ (or equivalently Z_1).



Values of $n_2^* \in [442, 884]$ that satisfy the CDL+Gao conditions are allowed.

At the final analysis, we reject H_0 if $Z_2 > 1.96$, where Z_2 is calculated without adjustment for adaptation.

Efficient sample size rules in the MP framework

We shall assess the conditional power that an increase in sample size achieves.

Suppose $Z_1 = z_1$ and we are considering a final sample size n_2^* with

$$Z_2(n_2^*) = \frac{\hat{\theta}(n_2)}{\sqrt{\{4\sigma^2/n_2\}}}$$

and conditional power under $\theta = \theta_0$

$$CP_{\theta_0}(z_1, n_2^*) = P_{\theta_0}\{Z_2(n_2^*) > 1.96 \mid Z_1 = z_1\}.$$

Setting γ as a “rate of exchange” between sample size and power, we shall:

Choose n_2^* to optimise a combined objective

$$CP_{\theta_0}(z_1, n_2^*) - \gamma(n_2^* - 442).$$

We specify $\theta = \theta_0$ as the effect size at which we are keen to “buy” additional power.

An overall optimality property

The rule that maximises $CP_{\theta_0}(z_1, n_2^*(z_1)) - \gamma n_2^*(z_1)$ for every z_1 also maximises, unconditionally,

$$P_{\theta=\theta_0}(\text{Reject } H_0) - \gamma E_{\theta_0}(N).$$

This can be seen by writing $P_{\theta=\theta_0}(\text{Reject } H_0) - \gamma E_{\theta_0}(N)$ as

$$\int \{CP_{\theta_0}(z_1, n_2^*(z_1)) - \gamma n_2^*(z_1)\} f_{\theta_0}(z_1) dz_1,$$

where $f_{\theta_0}(z_1)$ denotes the density of Z_1 under $\theta = \theta_0$, and noting that we have minimised the integrand for each z_1 .

So, the resulting procedure will have minimum possible $E_{\theta=\theta_0}(N)$ among all designs following the CDL+Gao framework that achieve the same power at $\theta = \theta_0$.

(5) Running R: **gain_adaptive**

Load the R function in the file **gain_adaptive.R** into your workspace.

Open the script **run_gain_adaptive.R** and use the commands there to run **gain_adaptive**.

You should see conditional power under effect size θ_0 plotted against final sample size n_2 , which takes values between $n_{2\min}$ and $n_{2\max}$.

The “combined objective” is this **conditional power** minus $(1/\gamma)(n_2 - n_{2\min})$.

We can interpret γ as before: the gain from a successful trial expressed as a multiple of the trial cost of 1 patient, discounted by a factor P_{sup} .

You can replace θ_0 , γ and $\hat{\theta}_1$ with your own values and run the script again.

Keeping values of θ_0 and γ fixed, notice how the optimal sample size varies with $\hat{\theta}_1$.

If you were sceptical about aspects of the problem formulation, do you see benefit in the consistency of sample size choices in this approach?

(5) Running R: **mp_opt_gain**

Load the R function in the file **mp_opt_gain.R** into your workspace.

Open the script **run_mp_opt_gain.R** and use it to run **mp_opt_gain**.

For selected values of γ , you should see plots of the sample size rule, where an optimum n_2 has been calculated for each value of $\hat{\theta}_1$.

The optimisation uses values of γ passed to the function **mp_opt_gain**.

The choice of n_2 in the range (442, 884) is constrained by the CDL+Gao conditions, which ensure the type I error rate is protected when a simple Z test is performed at the end of the trial.

Power and $E_{\theta}(N)$ curves are also drawn for the optimised adaptive designs.

Experiment to find a γ which gives similar power to the MP sample size rule.

By how much has the $E_{\theta}(N)$ curve been reduced?

How satisfactory is this — and what might we try next?

A subtle point

Suppose we have a prior probability P_{sup} that $\theta = \theta_0$, rather than $\theta = 0$.

Then, at an interim analysis we should really multiply the conditional power under $\theta = \theta_0$ by the conditional probability

$$P\{\theta = \theta_0 \mid Z_1 = z_1\},$$

rather than by P_{sup} .

For now, we can keep matters simple by supposing $P_{\text{sup}} = 1$.

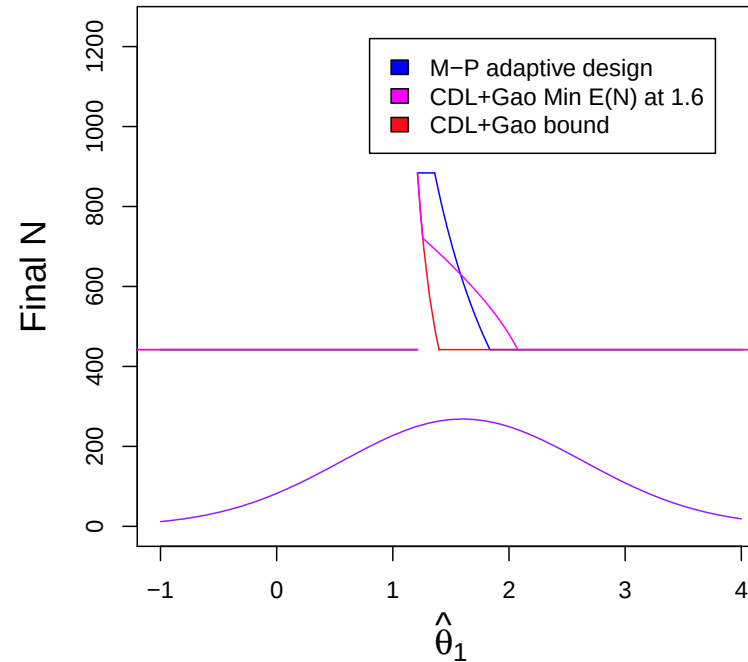
NB This still gives the optimality property (Slide 25) of minimising $E_{\theta=\theta_0}(N)$ among designs in the CDL+Gao framework that have the same power at $\theta = \theta_0$.

A Bayesian approach that factors $P\{\theta = \theta_0 \mid Z_1 = z_1\}$ into the conditional power calculation would have a similar property, minimising

$$(1 - P_{\text{sup}}) E_{\theta=\theta_0}(N) + P_{\text{sup}} E_{\theta=\theta_0}(N).$$

How does this compare with simply assuming $\theta = \hat{\theta}_1$?

Optimal sample size rule for $\theta_0 = 1.6$ and $\gamma = 0.000622$



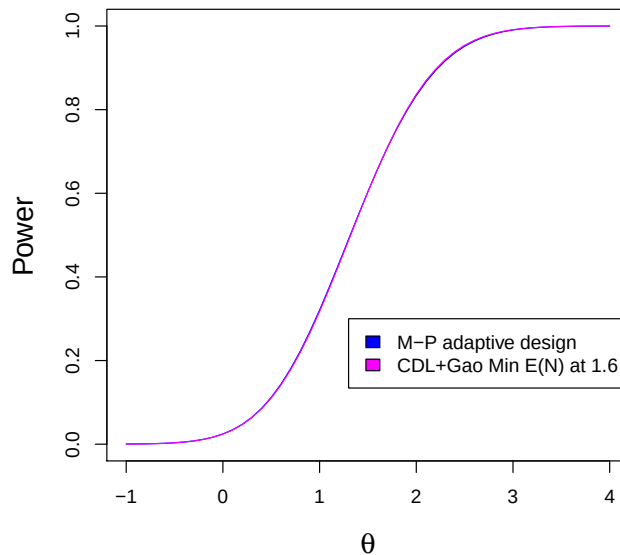
This rule gives power 0.658 at $\theta = 1.6$, the same as the MP design.

Decisions about the final sample size are based on a consistent comparison of the value of higher power and the cost of additional observations.

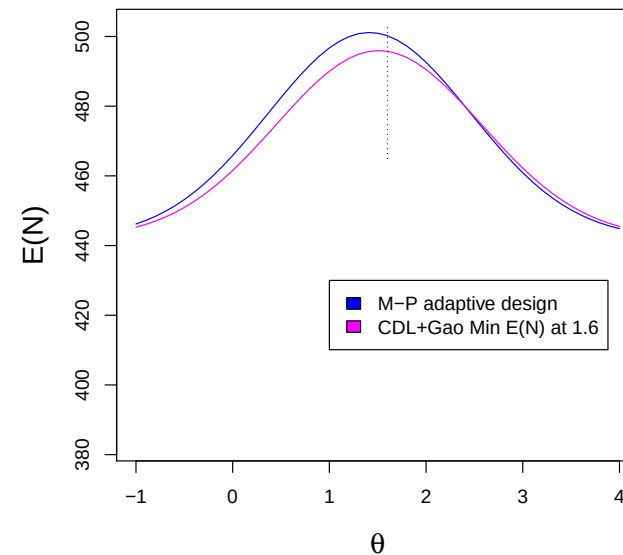
As $\hat{\theta}_1$ decreases, sample size increases less steeply than for the MP design.

Efficient sample size rules in the MP framework

Power curves



$E_{\theta}(N)$ curves



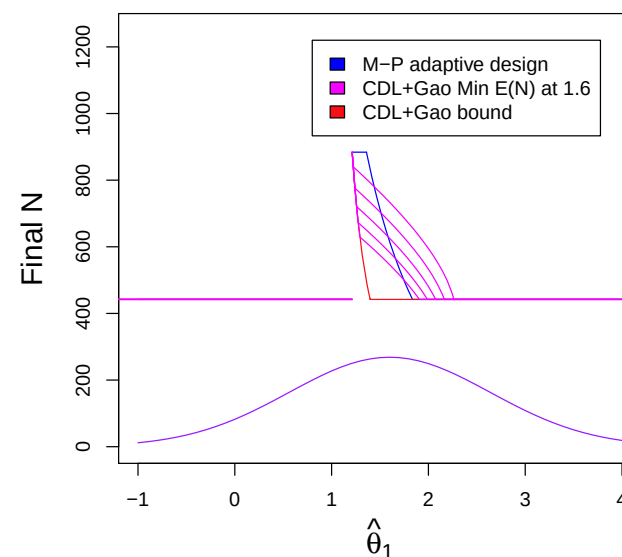
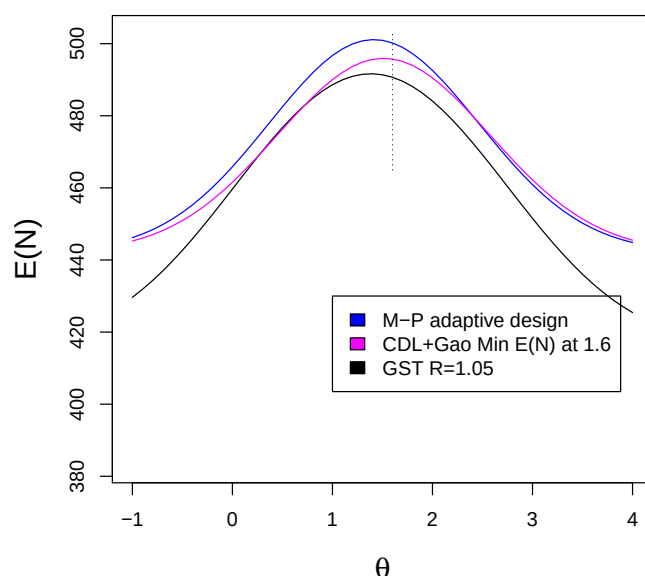
With the type I error rate at $\theta = 0$ fixed at 0.025, matching the MP design's power at one value of θ implies matching the whole power curve.

Our optimised design has the same power curve as the MP design and lower $E_{\theta}(N)$ (just about) at all θ values.

The reductions in $E_{\theta}(N)$ are modest — but given the optimality property of the sampling rule in the Mehta & Pocock framework, this is as good as it gets.

Further efficiency gains

Our new, optimised procedure still has higher $E_{\theta}(N)$ than a two-stage GST that ignores (but is charged for) pipeline data.



Shapes of optimised sample size rules suggest it would help to increase n_2^* at lower values of $\hat{\theta}_1$ — but this is not permitted in the CDL+Gao framework.

The **Conditional Probability of Rejection** principle, or equivalently using a Bauer & Köhne (*Biometrics*, 1994) **Combination Test** does allow such adaptations.

6. Using the Conditional Probability of Rejection principle

Reference: Proschan & Hunsberger, (*Biometrics*, 1995)

On observing $\hat{\theta}_1$, choose a new final sample size n_2^* .

Then, set the critical value for $Z_2(n_2^*)$ at the final analysis to maintain the Conditional Probability of Rejection (CPR) under $\theta = 0$ in the original design.

The overall type I error rate is the integral of the conditional type I error rate, and this remains the same.

This type of adaptation can also be regarded as a “weighted inverse normal combination test” Bauer & Köhne (1994).

We can follow our previous strategy in this new framework and set n_2^* to maximise $CP_{\theta_0}(z_1, n_2^*) - \gamma(n_2^* - 442)$, where θ_0 is an effect size of interest.

The resulting design has the minimum value of $E_{\theta_0}(N)$ among all designs in this larger class that achieve the same power under $\theta = \theta_0$.

(6) Running R: **cpr_opt_gain**

Load the R function in the file **cpr_opt_gain.R** into your workspace.

Open the script **run_cpr_opt_gain.R** and use it to run **cpr_opt_gain**.

For selected values of γ , you should see plots of the sample size rule, where an optimum n_2 has been calculated for each value of $\hat{\theta}_1$.

The optimisation uses values of γ passed to the function **cpr_opt_gain**.

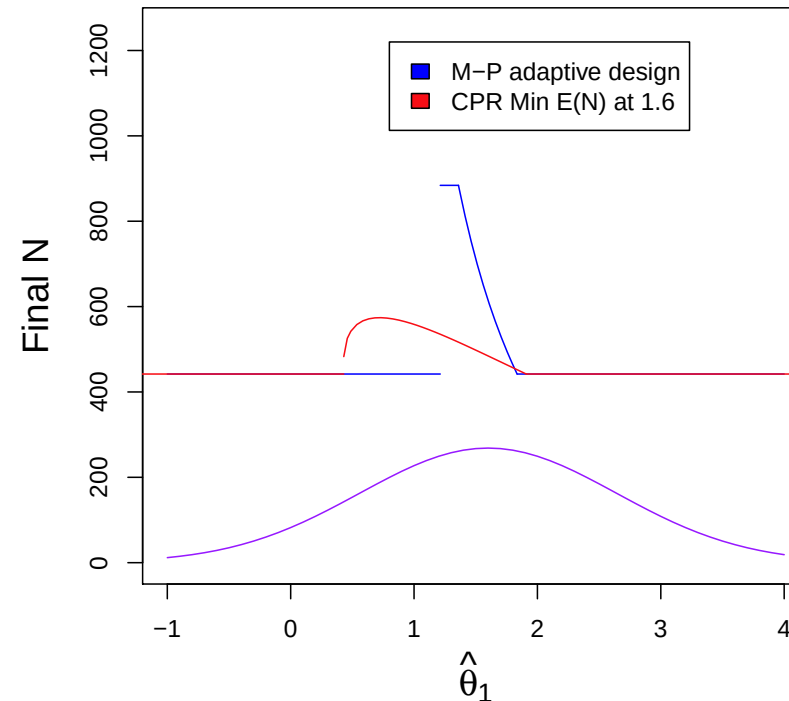
The choice of n_2 in the range (442, 884) is no longer constrained by the CDL+Gao conditions, since the type I error rate is protected by using a combination test.

Power and $E_{\theta}(N)$ curves are also drawn for the optimised adaptive designs.

Experiment to find a γ which gives similar power to the MP sample size rule.

By how much has the $E_{\theta}(N)$ curve been reduced now?

Optimal sample size rule for a CPR design with $\theta_0 = 1.6$



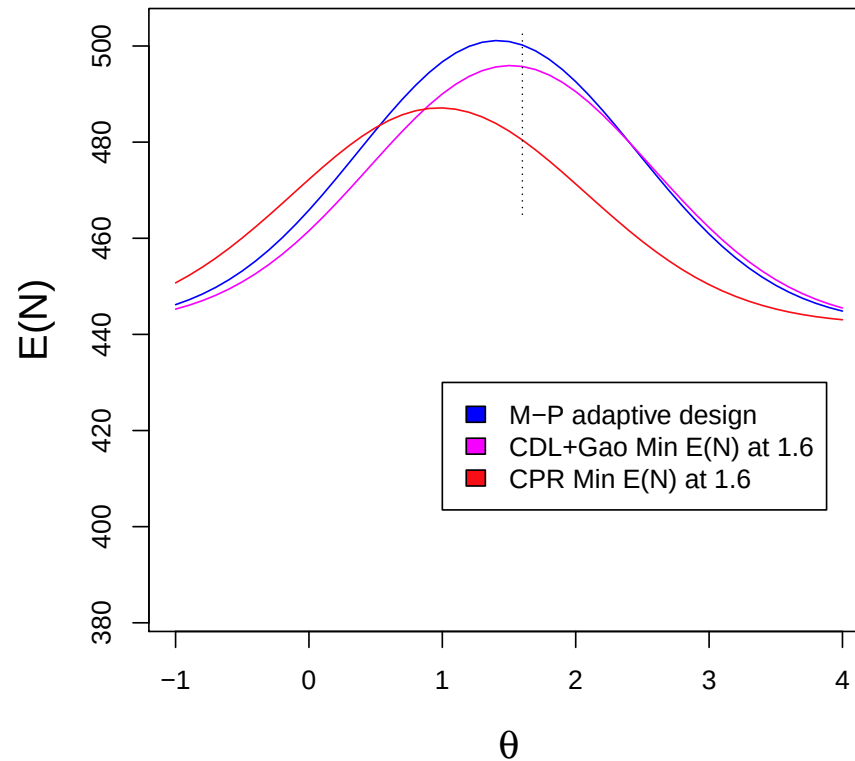
The rule with $\gamma = 0.00111$ matches the MP test's power of 0.658 at $\theta = 1.6$.

Shapes of optimised sample size rules are *very different* from the MP design.

The best opportunities for investing additional resource are *not* in Mehta & Pocock's "promising zone".

Efficient sample size rules in the CPR framework

$E_\theta(N)$ curves



The CPR principle allows sample size increases for $\hat{\theta}_1$ below the CDL+Gao region.

This leads to a useful reduction in $E_\theta(N)$ at $\theta = 1.6$.

7. Further options

The methods we have been working with can be generalised yet further.

1. In a more clearly Bayesian formulation, we would place a prior on θ and define an overall objective by weighting the power curve and the $E_{\theta}(N)$ function to reflect both prior beliefs and the importance of particular attributes.

2. Constraints on sample size can be tightened, setting the minimum as the current observations plus “pipeline” data — then n_2 could be as low as 416, rather than 442, in the M-P example (see `cpr2_opt_gain`).

Our optimised procedures have low maximum sample sizes (that is all they need to match the power of the MP design) — this could be increased.

3. There may be organisational benefits in fixing a discrete set of sample sizes. Most simply, one decides whether to add an extra group of a specified size: this relates to the “Delayed Response Group Sequential Designs” of Hampson & Jennison (2013), which extend nicely to the case of more than two analyses.